

Assessment Test: Software Tester (QA)

Candidate Details

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1. Test Case Design

Application Description

The application is a Task Management Application. It provides User Registration, User Login, and Create/View/Edit/Delete for tasks. Tasks are stored in a database and shown as a list.

Testing Assumption

The application is close to release and there's no existing test documentation, and no build was available to try things out on directly. A few details aren't mentioned in the requirements (password rules, field limits, whether there's a confirm dialog, session timeout, etc). Where a test case depends on one of these, I've noted it as something to confirm rather than assumed it as fixed behaviour.

2. Registration Test Cases

2.1 Positive

ID	Scenario	Steps / Input	Expected Result
REG-001	Register with valid details	Enter a valid name, a new email, and a password. Submit the form.	Should register successfully and show some confirmation (exact message/redirect to be confirmed).
REG-002	Login right after registering	Register, then try logging in with the same email/password.	Should be able to log in with the details just used to register.
REG-003	Register with minimum valid input	Use the shortest name/password that should still be acceptable.	Should register successfully if it meets whatever the minimum rule is.

2.2 Negative

ID	Scenario	Steps / Input	Expected Result
REG-004	Invalid email format	Enter an email without "@", e.g. riya.sharma.com.	Should show a validation error and not create the account.
REG-005	Missing required field	Leave the email (or name) field empty and submit.	Should show a "required" type error, form should not submit.
REG-006	Duplicate email	Try registering again with an email that's already used.	Should show an error that the email is already registered.

2.3 Edge Cases

ID	Scenario	Steps / Input	Expected Result
REG-007	Very long input in name/email	Enter an unusually long string in the name or email field.	Should not crash. (Exact max length isn't specified needs to be confirmed.)
REG-008	Password / confirm password mismatch	Enter different values in Password and Confirm Password, if that field exists.	Should show a mismatch error, assuming a confirm-password field is part of the form.
REG-009	Special characters in name	Enter something like <code><script>alert(1)</script></code> or symbols in the name field.	Should not break the page or run any script should just be treated as plain text.

3. Login Test Cases

3.1 Positive

ID	Scenario	Steps / Input	Expected Result
LOGIN-001	Login with valid details	Enter a correct, registered email and password.	Should log in and take the user to the task list/dashboard.

ID	Scenario	Steps / Input	Expected Result
LOGIN-002	Stay logged in while browsing	Log in, then move between task list and other pages.	Should not get logged out unexpectedly.

3.2 Negative

ID	Scenario	Steps / Input	Expected Result
LOGIN-003	Wrong password	Enter a correct email with an incorrect password.	Should show an error and not log the user in.
LOGIN-004	Unregistered email	Enter an email that was never registered.	Should show an error (ideally without confirming whether the email exists).
LOGIN-005	Empty fields	Leave email and password blank and submit.	Should show required-field errors.

3.3 Edge Cases

ID	Scenario	Steps / Input	Expected Result
LOGIN-006	Repeated failed logins	Enter the wrong password several times in a row.	Not sure if there's a lockout/CAPTCHA after repeated attempts this needs to be checked, since the requirements don't mention it.
LOGIN-007	Different letter case in email	Register with mixed-case email, then try logging in using lowercase.	Behaviour should be consistent either way to be confirmed.

4. Task CRUD Test Cases

4.1 Create Task

Positive

ID	Scenario	Steps / Input	Expected Result
CREAT E-001	Create task with valid data	Enter a title and description, then save.	Task should be saved and show up in the list.
CREAT E-002	Create task with only a title	Enter just a title, leave other fields blank, save.	Should still save if title is the only required field (needs to be confirmed).

Negative

ID	Scenario	Steps / Input	Expected Result
CREAT E-003	Empty title	Leave the title blank and try to save.	Should show an error and not save the task.
CREAT E-004	Very long title	Enter an unusually long string as the title.	Should not crash; exact limit isn't specified, so this needs checking.

Edge Cases

ID	Scenario	Steps / Input	Expected Result
CREAT E-005	Special characters/emoji in title	Enter symbols or an emoji in the task title.	Should save and display without breaking the page.
CREAT E-006	Double-click Save	Click Save twice quickly while creating a task.	Should not create two copies of the same task.

4.2 View Task List

Positive

ID	Scenario	Steps / Input	Expected Result
VIEW-001	View task list	Log in as a user who has tasks and open the list.	Should show that user's tasks correctly.
VIEW-002	List reflects changes	Add or edit a task, then check the list.	New/updated task should appear without needing a manual refresh.

Negative

ID	Scenario	Steps / Input	Expected Result
VIEW-003	Access another user's task	Try opening a task that belongs to a different user (e.g. by changing an ID in the link).	Should not be allowed this is worth checking carefully since it's not stated how task access is restricted.
VIEW-004	View list after session ends	Let the session expire (or log out), then try to open the task list.	Should ask the user to log in again rather than showing old data.

Edge Cases

ID	Scenario	Steps / Input	Expected Result
VIEW-005	No tasks yet	Log in as a new user with no tasks and open the list.	Should show some kind of "no tasks" message instead of a blank/broken page.
VIEW-006	Large number of tasks	Open the list for a user with a lot of saved tasks.	Should still load in reasonable time (performance not specified, worth a quick check).

4.3 Edit Task

Positive

ID	Scenario	Steps / Input	Expected Result
EDIT-001	Edit task with valid data	Change the title of an existing task and save.	Updated title should be saved and shown in the list.
EDIT-002	Edit one field only	Update only the description, leave the rest as is, save.	Only that field should change; the rest should stay the same.

Negative

ID	Scenario	Steps / Input	Expected Result
EDIT-003	Edit with invalid data	Clear the title field and try to save.	Should show an error and not save the change.
EDIT-004	Edit another user's task	Try editing a task that belongs to another user (e.g. by changing the ID).	Should not be allowed.

Edge Cases

ID	Scenario	Steps / Input	Expected Result
EDIT-005	Edit an already-deleted task	Delete a task in one tab, then try editing it from another tab, and save.	Should show a proper error (like "task not found") instead of crashing.
EDIT-006	Cancel an edit	Change some fields, then click Cancel instead of Save.	Changes should be discarded.

4.4 Delete Task

Positive

ID	Scenario	Steps / Input	Expected Result
DELET E-001	Delete a task	Click delete on a task and confirm.	Task should be removed from the list.
DELET E-002	Cancel a delete	Click delete, then choose Cancel if a confirmation shows up.	Task should remain if there's a confirmation step and it's cancelled.

Negative

ID	Scenario	Steps / Input	Expected Result
DELET E-003	Delete another user's task	Try deleting a task that belongs to another user (e.g. by changing the ID).	Should not be allowed.
DELET E-004	Delete without confirmation	Try sending a delete action directly without going through the normal confirmation step (if one exists).	Should still check permissions before deleting anything.

Edge Cases

ID	Scenario	Steps / Input	Expected Result
DELET E-005	Delete the same task twice quickly	Click delete twice fast, or try deleting from two tabs at once.	Second attempt should fail gracefully, not throw an error to the user.
DELET E-006	Delete the last task	Delete the only task remaining in the list.	List should then show the empty state correctly.

5. Input Validation & Error Handling

ID	Area	Scenario	Input / Action	Expected Result
VAL-001	Registration	Script tag in name field	<script>alert('x')</script>	Should be handled safely, no script should run.
VAL-002	Registration	SQL-like text in any field	'; DROP TABLE users;--	Should not cause a database error or break the form.
VAL-003	Login	SQL-like text in login fields	' OR '1'='1	Login should still fail normally, no bypass.
VAL-004	Login	Extra spaces around the email	" user@test.com "	Should probably be trimmed before checking against the stored email (to confirm).
VAL-005	Create Task	Extra spaces around the title	" Prepare report "	Should be trimmed before saving.
VAL-006	Edit Task	Non-English text/emoji in title	Title with Hindi text and an emoji	Should display correctly, not as broken characters.
VAL-007	Delete Task	Network drops mid-delete	Turn off network right after clicking delete	Should show a clear error rather than leaving things in an unclear state.
VAL-008	General	Check wording of error messages across forms	Trigger any validation error	Messages should be specific to the field, not a generic "error occurred".

6. Potential Bugs / Risk Areas

Identified by reading through the requirements, without running the actual application.

#	Bug Description	Severity	Reason / Impact
1	No mention of password rules for registration weak passwords may be accepted.	Major	Makes accounts easier to guess or break into.
2	Email check might be case-sensitive, so "Test@mail.com" and "test@mail.com" could both register.	Major	Could create duplicate accounts and confuse users.
3	No mention of a limit on repeated wrong login attempts.	Critical	Someone could keep guessing a password with no restriction.
4	Task IDs might be visible in the edit/delete links and possibly guessable.	Critical	Could let a user reach another user's task if IDs aren't checked properly.
5	Not clear if delete always asks for confirmation, or if that can be skipped.	Major	Risk of accidentally deleting a task with no way to undo it.
6	Not clear if validation is also checked on the server, not just on the screen.	Critical	If only the screen checks input, bad data could still get saved by going around the UI.
7	Not clear how long a login session lasts or what happens on logout.	Major	Sessions that don't expire are risky on shared computers.
8	No mention of how the task list behaves with a large number of tasks.	Minor	Could become slow to load once a user has many tasks.
9	Login error messages might differ for "wrong password" vs "email not found".	Minor	Could let someone figure out which emails are registered.
10	Not clear what happens if two tabs/sessions edit the same task at once.	Minor	One person's changes could get overwritten without any warning.

7. Testing Approach / Summary

Testing Approach

I went through each feature and first wrote the normal, expected-to-work case (positive), then thought about what happens with wrong or missing input (negative), and added a few edge cases like special characters, long input, and doing the same action twice quickly. Input validation is listed separately since a few checks like blocking script tags apply to more than one form.

A few details weren't in the requirements (password rules, field limits, session timeout, whether delete needs confirmation, etc). For those, I've written the test case around what should probably happen, but marked it as something to confirm rather than treating it as a fixed rule.

Key Risks Identified

- Whether a user can access or change another user's task by changing the task ID not addressed in the requirements, so worth checking early.
- Whether there's any limit on repeated wrong login attempts.
- Whether input is checked again on the server side, not just on the screen.

These three would be my priority to verify first, since they affect whether other users' data is safe, not just whether a form behaves as expected.

8. Conclusion

These test cases cover the main features listed registration, login, and task create/view/edit/delete along with basic input checks. Some of the expected results depend on rules that weren't given in the requirements, so those would need to be confirmed against the actual application before this is treated as final. This isn't a complete test suite, but it should be a reasonable first pass given only the feature list to work from.